Repeat Viewing Dominates: Graph, Factorization and Learned Re-ranking for TV Programme Recommendation on sweet.tv

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Code: https://github.com/shavrinaanastasia/sweettv-recsys-exam

Abstract — We predict, for 1,260 sweet.tv users, the five TV programmes they will watch most often in weeks 31-42, given a channel-level view log (weeks 11-30) in which only about 11% of schedule rows carry a programme id. We join views to labelled airings with the host's own "≥ 80% watched" rule, compare a popularity baseline with implicit ALS, content-based filtering and personalized PageRank on one temporal hold-out, and combine them by score fusion followed by a LightGBM LambdaRank re-ranker. The final strategy reaches MAP@5 = 0.352 on untouched validation users (popularity: 0.224) and a returned Kaggle private score of 0.43344 (popularity: 0.28773). The main lesson is that this target rewards predicting repeat viewing on a user's habitual channels; content similarity alone does not even beat popularity.

Index Terms — implicit feedback, personalized PageRank, matrix factorization, learning to rank, TV recommendation

1. Introduction

Problem. sweet.tv logs which channel each user watched and when, and publishes the broadcast grid separately. For each of 1,260 test users we must output five programme ids, ordered, that best match the five programmes the user will watch most often (≥ 80% of an airing) in the next 12 weeks; the metric is MAP@5 [1].

Motivation. A TV service's home screen has room for a handful of tiles. If they reflect what a viewer will really come back to (the evening news, a children's series) the screen saves effort; if not, it is ignored. The data also forces two non-trivial steps: views must first be turned into user-programme interactions, and every method must extrapolate forward in time.

Concrete example. User 8377619604347126107 watched 179 airings to ≥ 80% in weeks 11-20, spread over 22 programmes. In weeks 21-30 their five most-watched programmes were 'ТСН. Неделя', 'Сегодня. Итоги', 'Факты. День', 'Завтрак с 1+1' and a 'Свинка Пеппа' series id — all five already in their history. Personalized PageRank returns exactly these five (AP@5 = 1.0, Table 4).

Contributions. (i) A reproducible join that applies the host's label recipe, and a validation set built with the same recipe. (ii) A like-for-like comparison of four methods plus a channel-exposure signal, and a two-stage hybrid whose gain over every single method holds both on held-out validation users and on the Kaggle private split.

2. Related Work

Prior work consulted:

- **Implicit-feedback matrix factorization** [2] turns counts into binary preference with a confidence weight and solves it by alternating least squares; our M1 implements those update equations directly.
- **Personalized / topic-sensitive PageRank** [3] biases the random-walk restart vector towards a chosen set of nodes; our M3 restarts on each user's own history over a programme co-view graph.
- **Content-based recommendation** [4] builds item vectors from attributes and a user profile from liked items; our M2 uses TF-IDF over category, genres, decade, director and actor tokens.
- **Learning to rank**: LambdaMART [7] optimises a ranking metric through gradient-boosted trees; we use the LightGBM implementation [6] for the re-ranking stage.
- The competition page [1] defines the data, the metric and the exact label recipe.

Alternatives considered. Association rules [5] were not implemented: with 6,000+ programmes and per-user baskets dominated by daily repeats, rules mostly restate "watches A → watches A again", which history already captures. Sequence models (session RNN / transformers) were not tried: views are channel intervals, not item clicks, and only 20 weeks of history exist. A two-tower model was not tried for the same reason and because the hand-built graph and factorization already give strong collaborative signals at a fraction of the cost.

3. Method

Approach. Turn the view log into (user, programme, number of airings watched ≥ 80%) interactions, score every candidate programme with each method, and rank. Every method shares one candidate pool, one fallback and one evaluation protocol. The final system uses the fixed-weight score fusion of all signals as a candidate generator (top 150 per user) and re-orders those candidates with a LambdaRank model.

Pipeline.

- *Join.* A view (user, channel, [start, stop]) is matched to a schedule row on the same channel whose airing [start, start + duration] overlaps it. Overlap seconds of all of the user's views are summed per airing and capped at the airing duration. *Watched* means overlap / duration ≥ 0.8 — exactly the host's label rule [1]. Fig. 1 justifies this cut on the data as well: screen time is U-shaped (zapping near 0, full watches at 1), and 0.8 sits in the flat valley, so small changes to it move few pairs. 20.4% of 3,014,129 user-airing pairs pass.

- `tv_show_id = 0`. Unlabelled airtime is never an item (the host removed it from the labels). It is used only as context: each user's viewing seconds per channel, over **all** airtime, feed the channel-exposure signal below.
- *Labels* (validation): per user, count airings watched ≥ 0.8 per programme in the target weeks, keep users with ≥ 5 such programmes, take the 5 most frequent (ties: more watched seconds).
- *Candidate pool:* labelled programmes that air in the target weeks — for the test, read from the weeks 31-42 *schedule* only. Re-recommending a programme the user already watched is **allowed for every method** (the target is repeat-heavy, and the host does not forbid it).
- *Fallback:* empty slots are filled from the popularity ranking, identically for every method. Each method turns its scores into a list by sorting its own scores inside the pool; scores are never compared across methods without per-user min-max normalisation.

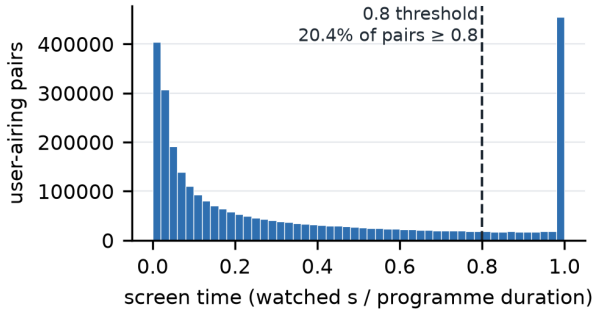


Fig. 1. Screen time of all 3,014,129 user-airing pairs (validation and test history). The 0.8 cut is the host's label rule.

Methods.

- *Popularity (baseline):* recency-weighted number of airings of the programme watched ≥ 0.8 by all users in the train weeks (half-life 4 weeks).
- *History (reference, not a counted method):* the user's own count of ≥ 0.8 airings per programme.
- *M1 implicit ALS* [2]: r = recency-decayed counts, confidence c = 1 + α r; 16 factors, λ = 30, α = 1, 10 iterations, seed 42.
- *M2 content-based* [4]: TF-IDF item vectors over 17,193 metadata tokens; user profile = count-weighted sum of watched items; cosine score.
- *M3 personalized PageRank* [3]: item graph with edge weight = number of users who watched both items ≥ 20 min; row-normalised; restart probability 0.3 to the user's normalised history; 20 power iterations.
- *Channel exposure (context signal):* Σ over channels of (user's share of viewing seconds on channel c) × (airings of the item on c in the target weeks).
- *Hybrid:* per-user min-max normalised scores, weighted sum; weights by coordinate ascent on validation half A: channel 2.0, PPR 1.5, history 0.75, ALS 0.5, CBF 0.1, popularity 0.05.
- *Final: hybrid + re-ranker:* top 150 fused candidates per user, re-ordered by LightGBM LambdaRank [6], [7] (300 trees, learning rate 0.05, 31 leaves, ≥ 100 rows per leaf, bagging 0.8, seed 42, deterministic) on 31 features: every component's raw score, normalised score and rank, fused score and rank, watch counts and seconds, target/train airing counts and their ratio, user activity. Relevance = 5 for the label's #1 down to 1 for #5.

Key design decisions. (1) *Use the host's 0.8 rule rather than an absolute-seconds cut,* because the score is computed on labels built with it; an absolute cut would mislabel long films and short news equally. (2) *Fit the re-ranker without validation labels:* for validation it learns history 11-15 → labels 16-20 and is then scored on 11-20 → 21-30; for the test it learns 11-20 → 21-30 (training data at that point) and is applied to 11-30 → 31-42. (3) *Keep PPR's full graph* (dense matrix product) after pruning to the 50 strongest edges per item cost up to 0.05 MAP (§5).

Tools & configuration. Python 3.11.9; numpy 2.4.6, pandas 3.1.0, scipy 1.17.1, pyarrow 26.0.0, LightGBM 4.7.0 (native `lgb.train` API), matplotlib 3.11.2, reportlab 5.1.0, Kaggle CLI 2.2.5. ALS and PPR are implemented from the equations in [2], [3] (`src/methods.py`), not from a library. Windows 11, 8 logical CPUs, 16 GB RAM, no GPU. Seeds: user A/B split 0, ALS 42, LightGBM 42. Runtime: parse 42 s, join 12 s; scoring the 1,260 test users: PPR 41 s, ALS 6 s, CBF 0.3 s (`results/predict_summary.json`); re-ranker test run 183 s end-to-end.

4. Experiments

Setup. View log: 5,821,987 views by 4,914 users on 202 channels (weeks 11-30); schedule: 381,892 labelled airings. Only 10.8% of schedule rows (16.0% of airtime) carry an id, yet 52.4% of all viewing seconds fall on that labelled airtime. *Validation split:* train weeks 11-20 → labels from weeks 21-30, giving 4,030 users with five labels (1,150 of them are Kaggle test users). The test target window is later than all training weeks, so this split preserves temporal order and has a similar history-to-horizon shape (10/10 weeks vs. 20/12 for the test). Users are split 50/50 at random (seed 0): every hyper-parameter and fusion weight is chosen on half A (2,048 users); half B (1,982 users) is reported untouched, mirroring the leaderboard's public/private user split [1]. Cold start: all 1,260 test users appear in the training log, so there is no user without any views; 22 of them have fewer than five programmes with ≥ 0.8 history, so their history-only list is topped up from the popularity fallback (`results/predict_summary.json`).

Table 1: MAP@5. Validation = train 11-20, labels 21-30; Kaggle = train 11-30, hidden labels 31-42. Sources: `results/hybrid_val.json`, `results/rerank_val_n150.json`, `results/kaggle_submissions.csv`.

Method	Val. A	Val. B (untouched)	Val. test users	Kaggle public	Kaggle private
Popularity (baseline)	0.2226	0.2244	0.2486	0.30344	0.28773
History (reference)	0.3105	0.3029	0.3468	—	—
M1 implicit ALS	0.2890	0.2862	0.3120	0.37750	0.36917
M2 content-based	0.1871	0.1831	0.2036	0.22711	0.21839
M3 personalized PageRank	0.3239	0.3182	0.3610	0.39959	0.38530
Channel exposure (signal)	0.3227	0.3192	0.3477	—	—
Hybrid, fixed weights	0.3484	0.3420	0.3810	0.44153	0.42921
Hybrid + re-ranker (final)	**0.3563**	**0.3524**	**0.3933**	**0.43978**	**0.43344**

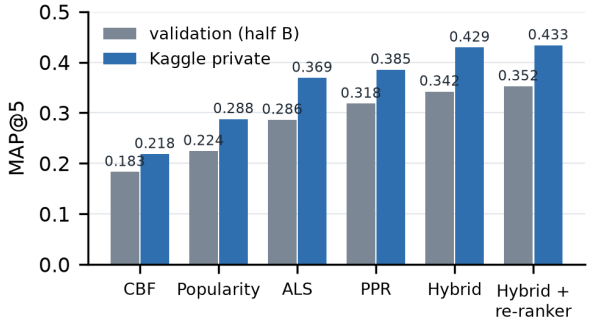


Fig. 2. Validation MAP@5 (half B) and returned Kaggle private MAP@5 per method. The order of the methods is the same on both.

Comparison vs. baseline. Every collaborative signal beats popularity by a wide margin: ALS +0.062 and PPR +0.094 on half B; the final strategy +0.128 on half B and +0.146 on Kaggle private. Content-based filtering is the only method below the baseline (−0.041 on B). The methods rank in the same order on validation and on Kaggle private (Fig. 2); Kaggle scores are higher overall, plausibly because test predictions use 20 instead of 10 weeks of history — a hypothesis we did not test.

Hybrid and ablation. Removing one component from the fixed-weight hybrid (half B, `results/hybrid_val_run1.json`): channel −0.019, PPR −0.004, history −0.002, CBF −0.001, popularity −0.000, ALS +0.000. A second weight search with a wider grid (up to 6.0) ended *lower* on half A (0.3477 vs. 0.3484) and was not used. The re-ranker was compared over two candidate depths and two model sizes on half A (150 candidates, small model best); recall of the label inside the 150 candidates is 0.767, which bounds what re-ranking can recover.

Tuning record. ALS took three grids: the best point sat on the grid edge twice ($\alpha = 5$, then $\lambda = 10$), so the grid was extended until an interior optimum appeared ($\lambda = 30$ beats 10 and 100; 16 factors beat 32 at $\lambda = 10$ and 8 at $\lambda = 30$). Weighting ALS by watched seconds instead of ≥ 0.8 counts collapsed MAP to 0.02-0.06, plausibly because short zapping visits then dominate the confidence matrix (not tested further). All grid points are logged in `results/val_runs.log` and `results/val_log_*.json`, including the losing ones.

Verification. (i) MAP@5 code reproduces the course slide's worked example (AP = 0.917 and 0.533). (ii) One user-airing overlap recomputed from the raw CSV with the standard `csv` module equals the join output (5,973 s of a 6,000 s film). (iii) All 4,030 validation label sets contain 5 distinct programmes, each watched ≥ 0.8 in weeks 21-30. (iv) ALS objective decreases every iteration (49,593 → 26,627 over 6 iterations). (v) PPR transition rows sum to 1, and each user's score vector keeps total mass 1. (vi) Each submission file has 1,260 rows, five distinct non-zero ids from the test-week pool, and was scored COMPLETE by Kaggle. (vii) The dense and sparse PPR implementations return the identical score (A = 0.32295) on the same graph. (viii) **[TODO: hand-compute AP@5 for the heavy viewer's popularity list in Table 4 and write the working here, e.g. hits at ranks 1, 3, 4, 5 → (1/1 + 2/3 + 3/4 + 4/5)/5 = 0.643]**. Sources: `results/checks.json`, `src/checks.py`.

4.1 Recommendation examples

Users were chosen by a fixed rule, not by inspecting results: among validation users who are also Kaggle test users, the one closest to the 90th percentile of training activity (heavy) and the one closest to the 10th percentile (light). Lists are from the validation models (train 11-20) so they can be compared with what the users actually watched in weeks 21-30. Titles are the schedule text of the programme's latest airing. One id can cover many airings with different titles: 12002955 appears under 207 episode titles of 'Свинка Пеппа', and 2400480 covers both the weekly 'ТСН. Неделя' and the daily 'ТСН.' news.

Table 4: Heavy viewer, user 8377619604347126107 (179 airings watched ≥ 80% in weeks 11-20, 22 distinct programmes). Bold = in the actual weeks 21-30 label; ⁿ = not watched ≥ 80% by this user in weeks 11-20.

Method	#1	#2	#3	#4	#5	AP@5
Actual 21-30	*2400480*	*2400508*	*240081*	*20088*	*12002955*	
Popularity	**2400480**	12001682	**240081**	**20088**	**2400508**	0.643
History (ref.)	**2400480**	**240081**	**12002955**	200338	**2400508**	0.760
M1 ALS	**2400480**	**20088**	**240081**	**12002955**	200344	0.800
M2 CBF	**2400480**	**240081**	2500261ⁿ	5900347ⁿ	2400467	0.400
M3 PPR	**2400480**	**240081**	**12002955**	**2400508**	**20088**	1.000
Hybrid (fixed)	**2400480**	**12002955**	**240081**	**20088**	12002896	0.800
Final: re-ranker	**2400480**	**240081**	**12002955**	**20088**	12002896	0.800

Table 5: Light viewer, user 2593914277176566399 (17 airings watched ≥ 80% in weeks 11-20, 13 distinct programmes). Bold = in the actual weeks 21-30 label; ⁿ = not watched ≥ 80% by this user in weeks 11-20.

Method	#1	#2	#3	#4	#5	AP@5
Actual 21-30	*20088*	*2400480*	*90077275*	*400455*	*90079862*	
Popularity	**2400480**	12001682ⁿ	240081ⁿ	**20088**	2400508	0.300
History (ref.)	6500479	**2400480**	90072234	**20088**	90057599	0.200
M1 ALS	**20088**	**2400480**	240081ⁿ	2400508	2400467ⁿ	0.400
M2 CBF	**2400480**	5900347ⁿ	2400467ⁿ	240081ⁿ	2500261ⁿ	0.200
M3 PPR	**2400480**	6500479	**20088**	400363	90021962	0.333
Hybrid (fixed)	**2400480**	**20088**	6500479	240081ⁿ	2400508	0.400
Final: re-ranker	**2400480**	**20088**	2400508	240081ⁿ	6500479	0.400

Table 6: Titles and categories of every programme id in Tables 4-5, looked up from the schedule.

tv_show_id	Title (from schedule)	Category
20088	'Завтрак с 1+1'. Информационно-развлекательная программа.	Развлечения
200338	Готовим вместе.	Познавательное
200344	'Чистоnews'.	Развлечения
240081	Факты. День.	Инфо
400363	От пацанки до панянки, 4 сезон, 15 эп.	Развлечения
400455	Любовь на выживание, 4 сезон, 5 эп.	Познавательное
2400467	'Викна-Новости'.	Инфо
2400480	'ТСН. Неделя'.	Инфо
2400508	Сегодня. Итоги с Олегом Панютой.	Инфо
2500261	Подробности недели.	Инфо
5900347	'Деньги'.	Инфо
6500479	'Мир наизнанку'.	Познавательное
12001682	М/с 'Губка Боб Квадратные Штаны', 89 с. 'Красти Губка'.	Детям
12002896	М/с 'Вспыш и чудо-машинки', 4 сезон, 5 с. 'Суперогромный приз'.	Детям
12002955	М/с 'Свинка Пеппа', 1 сезон, 11 с. 'Икота'.	Детям
90021962	Х/ф 'Эйс Вентура: Когда зовёт природа'.	Фильмы
90057599	Х/ф 'Ангелы Чарли 2: Только вперед'.	Фильмы
90072234	Х/ф 'Социальная сеть'.	Фильмы
90077275	Х/ф 'Миллион способов потерять голову'.	Фильмы
90079862	Х/ф 'Глубоководный горизонт'.	Фильмы

4.2 Kaggle submission

[TODO: insert the screenshot of the Kaggle Submissions page here (private scores and name visible) — replace this line with ![Fig. 3. Returned Kaggle scores; designated final submission: submission_rerank.csv, private 0.43344.](kaggle_screenshot.png){w=16}]

The designated final submission is `submissions/submission_rerank.csv` (ref 57803755), chosen on validation before its Kaggle score was known.

5. Discussion

[TODO: §5 must be in your own words (course rule §7.3, step 4). The text below is an AI draft based on the results — rewrite it, keep only claims you agree with, then delete this marker.]

5.1 Which recommendation strategy would we select for sweet.tv, and why?

Familiar vs. new. For the heavy viewer every method's hits are programmes already in their history (Table 4); the only "new" items come from CBF, and both are misses. For the light viewer two of the five actual programmes are films they had never watched (90077275, 90079862): no method predicts them, because a one-off film leaves no repeat signal and the co-view graph only knows past airings. On this target the useful lists are mostly *familiar*; an unfamiliar item is not evidence of a good discovery, and the data cannot tell whether the user enjoyed anything they watched.

How the methods differ. Popularity already does well because the top of its ranking (2400480 'ТСН', 20088 'Завтрак с 1+1') sits in both users' labels — it scores 0.643 for the heavy viewer without any personal data. ALS recovers the heavy

viewer's four main habits but fills slot 5 with 200344 instead of 2400508, and for the light viewer adds two popular news programmes they had never watched (240081, 2400467): low-rank smoothing pulls towards what similar users watch. PPR's diffusion stays close to the restart set, so it reorders the user's own programmes by how central they are in the co-view graph — which ranked the heavy viewer's five habits perfectly — and for the light viewer it keeps their own entertainment show and film (400363, 90021962), which ALS and popularity drop. CBF is meant to help where co-view signal is thin, but here it returns Инфо programmes for both users: attributes describe *what* a show is, not *how often* it airs or is watched, so CBF scores below popularity. Association rules would mostly repeat the history signal (§2). The strongest single cue is channel exposure: viewers return to the same channels, and a programme that airs often on those channels in the coming weeks is likely to be counted.

Why the final strategy. It is the best on untouched validation users (0.352 vs. 0.318 for PPR) and on Kaggle private (0.433 vs. 0.385). Cost is modest at this scale: one dense 9,671 × 9,671 matrix product per PPR step, a 16-factor ALS, and a re-ranker over 150 candidates per user — the whole test run takes about three minutes on an 8-core CPU, and would need re-engineering (sparse top-k graph, approximate PPR) at a much larger catalogue, at the accuracy cost shown in §5.2. For the user, the lists are reliable "continue watching / your channels" rows; they are not a discovery engine.

Limitations. Only 10.8% of schedule rows carry an id, so roughly half of all viewing seconds (47.6%) never becomes an interaction; the model is blind to whatever users watch on unlabelled channels. Ids mix series-level and episode-level programmes (Table 6), and the host's tie-breaking inside the label is unspecified. Absence from the log does not mean a user never saw a programme — the log covers this platform only. The re-ranker's private gain over the fixed hybrid is small (+0.004), and on the public half it is slightly lower (−0.002), so its advantage beyond validation is not established.

5.2 Failure case, root cause, fix

Failure. The first full validation run of PPR took 452.8 s for a single grid point (`results/val_run1_stopped.log`), so the planned 8-point grid would have taken about an hour. **First fix and its failure.** Pruning the graph to each item's 50 strongest edges cut the time to 38 s but dropped MAP on half A from 0.323 to 0.277 (`results/val_runs.log`). **Root cause.** The co-view graph is nearly dense: among the 2,902 programmes with any viewer in weeks 11-20, 71% of all pairs share at least one viewer, so the scipy sparse product was effectively dense and slow; pruning removed many weak edges into popular programmes, and those edges were what pulled every walk towards the popular shows that dominate the labels. **Fix and confirmation.** Keep the full graph as a dense float32 matrix and use a BLAS matrix product: 141 s per point with exactly the same score (A = 0.32295) as the slow sparse version, so the full grid ran. A second, smaller failure: the first re-ranker run crashed because LightGBM's scikit-learn wrapper requires scikit-learn (`results/rerank_val_console_failed_sklearn.log`); we switched to the native `lgb.train` API rather than add a dependency.

5.3 What worked, what surprised us, next step

- *Worked:* building validation labels with the host's recipe — the method ranking on validation matched the Kaggle private ranking exactly.
- *Worked:* using unlabelled airtime as channel context instead of discarding it — the single largest ablation drop.
- *Surprised:* content-based filtering lost to popularity; ALS needed very strong regularisation (λ = 30) and few factors to be competitive.
- *Surprised:* Kaggle private scores were 0.035-0.087 higher than validation half B for every method.
- *Next:* add the re-ranker's candidates from the user's habitual channels directly (recall is capped at 0.767) and test whether the gap to Kaggle comes from the longer history by validating with a 20-week history window.

6. AI Usage Disclosure

- **Tool:** Claude Code in the Claude desktop app, model Claude Opus 5.5 (`claude-opus-5-5`); one continuous session, exported unmodified as the attached session log.
- **How AI was used:** reading the exam brief, slides and course guidelines (docs); planning; writing all code in `src/` (code generation); running every experiment and Kaggle submission from my approval; debugging (PPR runtime, LightGBM wrapper error, memory blow-up in a feature); searching and checking references (search); figures (viz); drafting this report and the README.
- **What I personally verified:** **[TODO: list only what you actually did yourself, e.g. "hand-computed AP@5 for the heavy viewer's popularity list (0.643, matches Table 4)", "re-ran src/checks.py and compared the output with results/checks.json", "opened the 7 reference links", "checked the Kaggle scores on the Submissions page against Table 1"]**.
- **What I trusted without verification:** **[TODO: be honest, e.g. LightGBM's LambdaRank implementation, the correctness of the TF-IDF construction, the parquet conversion beyond the row counts]**.
- **Reference check:** every reference was checked against Crossref metadata or the publisher's page on 2026-10-30 (`notes/reference_check.md`); the 1999 PageRank technical report was dropped because its server returned an error.
- **Session log:** the attached session export is the audit trail for all of the above.

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